

3.2)

$$a) (\vec{v}_1, \vec{v}_2, \vec{v}_3) = (1, x, x^2)$$

~~Pr~~ Provedeme G-S ortogonalizaci

$$i) \|\vec{v}_1\| = \sqrt{\int_1^5 1 \cdot 1 dx} = \sqrt{4} = 2$$

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{2}$$

$$ii) \vec{v}_2 - \vec{w}_2 = \vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1 = x - \left(\int_1^5 \frac{1}{2} x dx \right) \cdot \frac{1}{2} =$$

$$= x - 6 \cdot \frac{1}{2} = x - 3$$

$$\|\vec{v}_2 - \vec{w}_2\| = \sqrt{\int_1^5 (x-3)^2 dx} = \sqrt{\frac{16}{3}} \doteq 2,309$$

$$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{\sqrt{3}}{4} (x-3) \doteq 0,433 (x-3)$$

$$iii) \vec{v}_3 - \vec{w}_3 = \vec{v}_3 - \langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 - \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2 =$$

$$= x^2 - \left(\int_1^5 \frac{1}{2} x^2 dx \right) \cdot \frac{1}{2} - \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 dx \right) \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = x^2 - \frac{31}{3} - 6(x-3) =$$

$$= x^2 - 6x + \frac{23}{3}$$

$$\|\vec{v}_3 - \vec{w}_3\| = \sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3} \right)^2 dx} = \sqrt{\frac{256}{45}}$$

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} = \frac{\sqrt{45}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \doteq 0,419 \left(x^2 - 6x + \frac{23}{3} \right)$$

Na výstupu G-S ortogonalizace jsme dostali

posloupanost $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$, která tvoří ortonormální
bázi podprostoru $LO\{1, x, x^2\}$

b) Posloupanost $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$ je ON báze podprostoru.

Ortogonalní projekce $\vec{w}$ vektoru $\vec{v} = \sin x$ na tento podprostor je tedy:

$$\begin{aligned}\vec{w} &= \langle \vec{u}_1, \vec{v} \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v} \rangle \vec{u}_2 + \langle \vec{u}_3, \vec{v} \rangle \vec{u}_3 = \\ &= \left(\int_1^5 \frac{1}{2} \sin x \, dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{9} (x-3) \sin x \, dx \right) \frac{\sqrt{3}}{9} (x-3) + \left(\int_1^5 \frac{\sqrt{45}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \sin x \, dx \right) \frac{\sqrt{45}}{16} \left(x^2 - 6x + \frac{23}{3} \right)\end{aligned}$$

$$\cancel{= 0,123} \doteq 0,064 - 0,647x + 1,940 - 0,053x^2 + 0,315x - 0,403 \doteq$$

$$\doteq -0,053x^2 - 0,332x + 1,601$$

$$c) \left\| \sin x - (-0,053x^2 - 0,332x + 1,601) \right\| =$$

$$\doteq \sqrt{\int_1^5 \left(\sin x - (-0,053x^2 - 0,332x + 1,601) \right)^2 dx} \doteq \sqrt{0,102} \doteq$$

$$\doteq \cancel{0,1} 0,319$$